Homework 7: Machine Learning on a Real Prediction Problem

SPY daily-data snapshot run 6 November 2026 | Latest close in sample: 5 November 2026

1. Dataset

The feature table has **3,711 rows and 8 columns**. Each row at date t uses only data through the close at t; the label is the return from t to t+1. I used returns over 1, 5, 20, and 60 days, 20-day realised volatility, price relative to a 50-day average, and volume relative to its 20-day average.

Columns: return_1, return_5, return_20, return_60, realised_vol_20, price_to_ma50, volume_to_avg20, next_day_return.

Date	r1	r5	r20	r60	vol20	MA50 rel.	vol. rel.	next return
2012-02-03	0.0140	0.0206	0.0508	0.0656	0.0055	0.0623	0.1354	-0.0007
2012-02-06	-0.0007	0.0234	0.0528	0.0514	0.0054	0.0590	-0.2276	0.0025
2012-02-07	0.0025	0.0264	0.0529	0.0944	0.0054	0.0587	-0.0404	0.0030

Manual check: the first row's features end on 3 February 2012; only its label uses the next close.

2. Time split

Set	Date range	Rows
Training	2012-02-03 to 2020-12-07	2,226
Validation	2020-12-08 to 2023-11-17	742
Test	2023-11-20 to 2026-11-05	743

The ranges are consecutive, time ordered, and non-overlapping; no rows were shuffled.

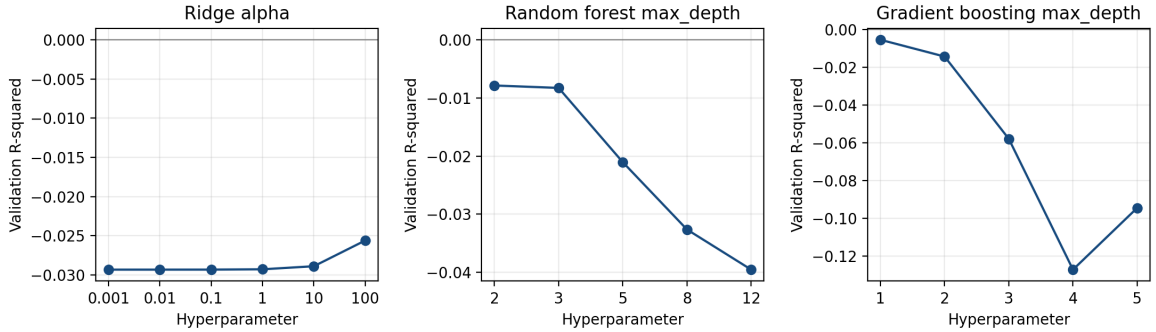
3. Initial model fit

Model	Training R-squared	Validation R-squared
Linear regression	+0.0299	-0.0293
Ridge (alpha=1)	+0.0299	-0.0293
Random forest (unrestricted)	+0.8438	-0.0562
Gradient boosting (depth=3)	+0.5797	-0.1102

The large train-validation gaps for the tree models are overfitting, not predictive power.

4. Validation-only tuning

Model / hyperparameter	Chosen value	Validation R-squared
Ridge alpha	100	-0.0256
Random forest max_depth	2	-0.0079
Gradient boosting max_depth	1	-0.0055



5. One held-out test

The shallow gradient booster had the highest validation score, even though it was still negative. It was selected before evaluating the test set.

Test metric	Result
Test R-squared	+0.0239
Correlation: prediction vs. actual next-day return	+0.1627

6. Test-set sign strategy

The rule holds SPY when the prediction is positive and otherwise holds cash. It uses the same close-to-next-close convention as Homework 4.

Metric	Sign strategy	Buy-and-hold
Annualised return	+18.45%	+17.84%
Maximum drawdown	-19.00%	-19.00%
Fraction invested	99.46%	100.00%

7. Gradient-boosting feature importances

Feature	Importance
return_1	0.541396
return_60	0.225347
price_to_ma50	0.112016
return_5	0.100778
realised_vol_20	0.016551
return_20	0.003912
volume_to_avg20	0.000000

8. Interpretation

Near-zero validation R-squared is the expected outcome here, not evidence of a coding bug. Daily SPY returns are extremely noisy, the simple price and volume features are public, and many sophisticated investors already test the same momentum, moving-average, volatility, and volume signals. Competition should eliminate an obvious, scalable next-day forecast; flexible trees can nevertheless fit noise in the training period, which is exactly what their large training-validation gaps show. The positive test R-squared is one holdout outcome after a negative validation result, so it is not enough evidence to claim an edge. A near-zero R-squared can still coexist with a profitable trading rule. R-squared measures squared-error accuracy for the size of every daily return, whereas a sign rule only needs a small, persistent advantage in conditional expected return or direction, potentially concentrated on a subset of days. But this particular strategy was invested 99.46% of the time and its small return advantage is consistent with being almost buy-and-hold; it does not establish a predictive edge, especially before transaction costs, turnover, and a genuinely prospective test.